Exp No:8. Experiment to understand Linear Regression for a given data set.

Code:

```
import numpy as np

import pandas as pd


df = pd.read_csv('Salary-data.csv')

df.info()

df.dropna(inplace=True)

df.info()

df.describe()


features = df.iloc[:, [0]].values

label = df.iloc[:, [1]].values


from sklearn.model_selection import train_test_split


x_train, x_test, y_train, y_test = train_test_split(

    features, label, test_size=0.2, random_state=1

)


from sklearn.linear_model import LinearRegression


model = LinearRegression()
```

```
model.fit(x_train, y_train)

model.score(x_train, y_train)
model.score(x_test, y_test)

model.coef_
model.intercept_

import pickle

pickle.dump(model, open('salarypred.model', 'wb'))

model = pickle.load(open('salarypred.model', 'rb'))

yr_of_exp = float(input("enter year"))
yr_of_exp_NP = np.array([[yr_of_exp]])
Salary = model.predict(yr_of_exp_NP)

print(
   "Estimated salary for {} year of experience is {}".format(
      yr_of_exp, Salary
   )
)
```

Output:

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

```
▾ LinearRegression
LinearRegression()
```

```
0.9603182547438908
```

```
array([[9281.30847068]])
```

```
array([27166.73682891])
```

Result:

Hence , This code has executed successfully . linear regression is verified.